Long Properly Colored Paths in the Union of Two Hamiltonian Paths: An Extremal Conjecture and Its Refutation

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Abstract

For a permutation $\sigma \in S_n$, let H_σ be the two-edge-colored multigraph on $\{0, 1, \ldots, n-1\}$ whose blue (P) edges form the natural-order Hamiltonian path and whose red (V) edges form the Hamiltonian path induced by the value order of σ. A properly colored (PC) path is a simple path whose consecutive edges alternate in color. Let $\rho(\sigma)$ be the maximum number of vertices on such a path, counted with the convention that a path on k vertices has $k-1$ edges. The parameter $\rho_{\min}(n) := \min_{\sigma \in S_n} \rho(\sigma)$ controls a matching-edge invariant in the framework of Norin, Steiner, Thomassé, and Wollan [2]: any linear lower bound $\rho_{\min}(n) \geq cn$ with $c > 0$ would push the best bound on the Lovász path problem from $\Omega(n^{9/14})$ to $\Omega(n^{2/3})$.

We prove a direct conversion lemma from PC paths in H_σ to matching-edge-rich paths in an associated graph $G(n, \sigma)$. We analyze a block-reversal family $\sigma_m \in S_{4m+3}$ for which an explicit PC path on $2m+3 = (n+3)/2$ vertices is given, together with a matching upper bound obtained from a local block calculus. The natural half-bound conjecture $\rho(\sigma) \geq \lceil (n+3)/2 \rceil$, supported by exhaustive enumeration through $n \leq 14$ and by the analysis of σ_m, turns out to be false: at $n = 18$, deleting one interior position of $\sigma_4 \in S_{19}$ gives $\sigma^* \in S_{18}$ with $\rho(\sigma^*) = 10 < 11$. A stackable variant gives exact computed values through four periods, with slope $5/11$ matching at every recorded k; whether this slope continues asymptotically (and thus whether $c_* := \liminf_n \rho_{\min}(n)/n \leq 5/11$) is conjectural. The unconditional bound $\rho(\sigma) \geq s_{\max}(\sigma)$ (the largest ± 1-block size) survives. Whether $c_* > 0$ remains open.

1 Background and motivation

Lovász [1] asked at the 1969 Calgary conference (published 1970) whether every connected vertex-transitive graph on n vertices contains a Hamiltonian path. The best general bound is $\Omega(n^{9/14})$, due to Norin, Steiner, Thomassé, and Wollan [2], and rests on a reduction that depends on the matching-edge content of long simple paths in a family of graphs $G(n, \sigma)$ on $2n$ vertices, indexed by $\sigma \in S_n$. The current bound $f(n) = \Omega(n^{4/5})$ on this auxiliary parameter comes from Liu, Yu, and Zhang's circumference theorem [3] on 3-connected cubic graphs; replacing it with $f(n) = \Omega(n)$ would yield $\Omega(n^{2/3})$ on the Lovász problem.

Specifically, define

$$f(n) \;=\; \min_{\sigma \in S_n} \max_{\pi} \mu(\pi),$$

where π ranges over simple paths in $G(n, \sigma)$ and $\mu(\pi)$ counts the matching edges used by π. This note converts the question on $G(n, \sigma)$ into a question about PC paths in a smaller two-edge-colored graph H_σ on n vertices, formulates the natural extremal conjecture, proves it for a

specific locally extremal family, and exhibits explicit counterexamples that refute the global form. The counterexamples extend to a stackable family for which the original asymptotic constant 1/2 cannot be correct.

The paper is organized as follows. Section 2 introduces H_σ and proves the Conversion Lemma. Section 3 states the half-bound conjecture and analyzes the block-reversal family σ_m. Section 4 presents the $n = 18$ counterexample and the stacked construction. Section 5 collects unconditional structural results that survive the refutation. Sections 6–8 discuss the landscape of the problem, the present status, and natural next questions.

Symbol	Meaning
σ	a permutation in S_n, viewed as a bijection $\{0,\dots,n-1\} \to \{0,\dots,n-1\}$
$G(n,\sigma)$	the matching-edge graph on $2n$ vertices (Definition 2.1)
H_σ	the two-edge-colored multigraph on n vertices (Definition 2.2)
$\rho(\sigma)$	max number of vertices on a PC path in H_σ
$\rho_{\min}(n)$	$\min_{\sigma\in S_n} \rho(\sigma)$
$f(n)$	$\min_\sigma \max_\pi \mu(\pi)$ in $G(n,\sigma)$
$b(\sigma)$	number of ± 1-agreement blocks of σ
$s_{\max}(\sigma)$	largest block size
$d(\sigma)$	deficit $n - \rho(\sigma)$
τ_σ	block-quotient permutation in $S_{b(\sigma)}$
c_*	$\liminf_n \rho_{\min}(n)/n$

Table 1: Notation used throughout.

Notation summary (Table 1).

2 The graph $G(n,\sigma)$ and the color reduction

Definition 2.1. For $\sigma \in S_n$, viewed as a bijection $\{0,\dots,n-1\} \to \{0,\dots,n-1\}$, let $G(n,\sigma)$ be the graph on $2n$ vertices $\{v_0,\dots,v_{n-1},u_0,\dots,u_{n-1}\}$ with edges:

- a top path P: $v_0 v_1 \cdots v_{n-1}$,
- a bottom path Q: $u_0 u_1 \cdots u_{n-1}$,
- a matching $M = \{v_i u_{\sigma(i)} : 0 \le i \le n-1\}$.

Each v_i is incident to exactly one matching edge, and each u_j is also incident to one (since σ is a bijection). The four path endpoints $v_0, v_{n-1}, u_0, u_{n-1}$ have degree 2 (one path edge and one matching edge each); every other vertex has degree 3 (two path edges and one matching edge). The parameter of interest is $f(n) = \min_\sigma \max_\pi \mu(\pi)$, where $\mu(\pi)$ counts the edges of π lying in M.

Definition 2.2. Let H_σ be the two-edge-colored multigraph on $\{0,\dots,n-1\}$ with edges

- P-edges (blue): $(i, i+1)$ for $i = 0,\dots,n-2$;
- V-edges (red): $(\sigma^{-1}(v), \sigma^{-1}(v+1))$ for $v = 0,\dots,n-2$.

A *properly colored* (PC) path in H_σ is a simple path whose consecutive edges alternate in color, and $\rho(\sigma)$ is the maximum number of vertices on such a path.

Convention 2.3. We count ρ in *vertices*, not edges. A PC path on k vertices has $k-1$ edges.

Remark 2.4 (Parallel edges). H_σ is a multigraph. The same unordered pair $\{i,j\}$ can carry both a P-edge and a V-edge; we treat these as parallel edges of different colors. A PC path may traverse such a pair only once and must commit to one color for the traversal. This doubled-edge phenomenon drives the block flexibility analyzed in Section 3.

Both color classes of H_σ are Hamiltonian paths on $\{0,\dots,n-1\}$: the P-class is the natural order, and the V-class is the order $\sigma^{-1}(0),\sigma^{-1}(1),\dots,\sigma^{-1}(n-1)$. Each vertex has color-P degree at most 2 and color-V degree at most 2, so the maximum degree in H_σ is 4. This places the problem outside the regimes of standard PC-path results, which typically require minimum degree at least linear in n [4].

A useful picture: plot σ as the point set $\{(i,\sigma(i)) : 0 \le i \le n-1\}$ in the plane. A P-edge connects horizontally adjacent columns; a V-edge connects vertically adjacent rows. A PC path traces an alternating horizontal/vertical staircase through this grid pattern.

2.1 The Conversion Lemma

The reduction from ρ to matching-edge content is exact and constructive.

Theorem 2.5 (Conversion Lemma)**.** *If H_σ has a PC path on k vertices, then $G(n,\sigma)$ has a simple path on $2k$ vertices that uses exactly k matching edges. Hence $\max_\pi \mu(\pi) \ge \rho(\sigma)$ for every σ, and taking the minimum over σ gives $f(n) \ge \rho_{\min}(n)$.*

Proof. Let $\pi = a_0a_1\cdots a_{k-1}$ be a PC path in H_σ with edge colors $c_0,c_1,\dots,c_{k-2}$ alternating. We construct a simple path Π in $G(n,\sigma)$ on the $2k$ vertices $\{v_{a_i} : 0 \le i < k\} \cup \{u_{\sigma(a_i)} : 0 \le i < k\}$.

Construction. Write M for the matching edge between a top vertex v_x and bottom vertex $u_{\sigma(x)}$. If c_0 is a P-edge, Π runs

$$u_{\sigma(a_0)} \xrightarrow{M} v_{a_0} \xrightarrow{P} v_{a_1} \xrightarrow{M} u_{\sigma(a_1)} \xrightarrow{Q} u_{\sigma(a_2)} \xrightarrow{M} v_{a_2} \xrightarrow{P} v_{a_3} \xrightarrow{M} \cdots$$

If c_0 is a V-edge, Π runs symmetrically starting at v_{a_0}.

Edge validity. A P-edge a_ia_{i+1} in H_σ means $|a_i - a_{i+1}| = 1$, so $v_{a_i}v_{a_{i+1}}$ is a top-path edge in $G(n,\sigma)$. A V-edge a_ia_{i+1} in H_σ means $|\sigma(a_i) - \sigma(a_{i+1})| = 1$, so $u_{\sigma(a_i)}u_{\sigma(a_{i+1})}$ is a bottom-path edge. Color alternation in H_σ matches the alternation of top vs. bottom traversal in Π, so every edge of Π exists in $G(n,\sigma)$.

Simplicity. The top vertices on Π are $\{v_{a_i}\}$, distinct because the a_i are distinct. The bottom vertices are $\{u_{\sigma(a_i)}\}$, distinct because σ is a bijection. The two families are disjoint.

Matching count. Π uses one matching edge per vertex of π, namely $v_{a_i}u_{\sigma(a_i)}$ for $i = 0,\dots,k-1$, for a total of k.

For each σ, the construction gives $\max_\pi \mu(\pi) \ge \rho(\sigma)$. Taking the minimum over σ gives $f(n) \ge \rho_{\min}(n)$. □

3 The half-bound conjecture and its tight family

Computation at $n \le 14$ and the structural analysis below pointed at the constant 1/2. The conjecture below was formulated on the basis of this evidence and is the headline question this paper revisits.

Conjecture 3.1 (Half-bound conjecture)**.** *For every $n \ge 2$ and every $\sigma \in S_n$,*

$$\rho(\sigma) \;\ge\; \left\lceil \tfrac{n+3}{2} \right\rceil.$$

Conjecture 3.1 is false; the counterexample is given in Section 4. We retain the statement because the locally extremal family that motivated it still saturates the bound at $n = 4m + 3$, and the structural lemmas developed for that family are the main positive results.

3.1 The block-reversal family

For $m \geq 1$ and $n = 4m + 3$, partition $\{0, 1, \ldots, n-1\}$ into consecutive blocks of sizes

$$(2, \quad \underbrace{3,\ 1,\ 3,\ 1,\ \ldots,\ 3,\ 1,\ 3}_{m \text{ threes alternating with } m-1 \text{ singletons}} \quad , 2),$$

and let σ_m be the permutation obtained by reversing each block. For $m = 1$ this collapses to the composition $(2, 3, 2)$ with no interior singletons. Explicitly:

$$\begin{aligned}
m = 1,\ n = 7: &\quad \sigma_1 = (1, 0, 4, 3, 2, 6, 5), \\
m = 2,\ n = 11: &\quad \sigma_2 = (1, 0, 4, 3, 2, 5, 8, 7, 6, 10, 9), \\
m = 3,\ n = 15: &\quad \sigma_3 = (1, 0, 4, 3, 2, 5, 8, 7, 6, 9, 12, 11, 10, 14, 13), \\
m = 4,\ n = 19: &\quad \sigma_4 = (1, 0, 4, 3, 2, 5, 8, 7, 6, 9, 12, 11, 10, 13, 16, 15, 14, 18, 17).
\end{aligned}$$

Theorem 3.2 (Extremal value)**.** *For every $m \geq 1$, $\rho(\sigma_m) = 2m + 3 = (n + 3)/2$.*

Lower bound: an explicit PC path

Define T_m as follows: start with $(4m - 2, 4m - 1, 4m)$; for $j = m, m - 1, \ldots, 2$, append the pair $(4j - 3, 4j - 4)$; finally append $(0, 1)$.

For $m = 1$ this yields the specialization $T_1 = (2, 3, 4, 0, 1)$ on 5 vertices. For higher m:

$$\begin{aligned}
T_2 &= (6, 7, 8, 5, 4, 0, 1), \\
T_3 &= (10, 11, 12, 9, 8, 5, 4, 0, 1), \\
T_4 &= (14, 15, 16, 13, 12, 9, 8, 5, 4, 0, 1).
\end{aligned}$$

The length of T_m is $3 + 2(m - 1) + 2 = 2m + 3$.

Why this is a PC path. The edges of T_m split into four groups:

- Internal edges within a ± 1-block, namely $(4m-2, 4m-1)$, $(4m-1, 4m)$, and $(0, 1)$: doubled, so freely colorable.
- Edges of the form $(4j,\ 4j - 3)$ for $j = m, m - 1, \ldots, 2$: each crosses from the rightmost vertex of a 3-block to the adjacent singleton. Gap is 3, so not a P-edge; direct verification on σ_m^{-1} shows V-only.
- Edges of the form $(4j - 3,\ 4j - 4)$ for $j = m, m - 1, \ldots, 2$: each crosses from a singleton to the rightmost vertex of the next 3-block. P-edge by adjacency, not V.
- The final cross-edge $(4, 0)$: from the leftmost 3-block to the leftmost 2-block. V-only.

Assigning colors $V, P, V, P, \ldots$ across the $2m + 2$ edges of T_m matches every forced color and uses the doubled edges in the remaining slots. The full color sequence is $V, P, V, P, \ldots, V, P$.

Local machinery for the upper bound

The upper bound $\rho(\sigma_m) \leq 2m+3$ uses a calculus of ± 1-blocks. The first three results (Definition 3.3, Lemma 3.4, Lemma 3.6, Lemma 3.7) apply to every σ; the rest are specific to σ_m.

Definition 3.3. A ± 1-*agreement block* of σ is a maximal interval $[p_\ell, p_r] \subseteq \{0, \ldots, n-1\}$ such that either $p_\ell = p_r$ (a *singleton* block), or there is a fixed $\varepsilon \in \{+1, -1\}$ with $\sigma(i+1) - \sigma(i) = \varepsilon$ for all $p_\ell \leq i < p_r$. The blocks partition $\{0, \ldots, n-1\}$. Write $b(\sigma)$ for the number of blocks and $s_{\max}(\sigma)$ for the largest block size.

Lemma 3.4 (Port Structure)**.** *Let $B = [p_\ell, p_r]$ be a block of size s in σ.*

(a) Every edge of H_σ internal to B (joining $i, i+1$ with $p_\ell \leq i < p_r$) is doubled: both P-colored and V-colored.

(b) Every edge of H_σ with exactly one endpoint in B is incident to p_ℓ or p_r. There are at most four such external edges, partitioned by orientation: P-prev, P-next, V-prev, V-next.

(c) If $s \geq 3$, every interior vertex of B has degree exactly 4 in H_σ, formed by two pairs of doubled edges.

Sketch. If $i, i+1 \in B$, then $|\sigma(i+1) - \sigma(i)| = 1$, so $\{i, i+1\}$ is both a P-edge and a V-edge. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ with exactly one endpoint in B must cross the boundary of $\sigma(B)$, which only happens at the ports. (c) follows from (a) at an interior position. □

Definition 3.5. Given a PC path P in H_σ and a block B, a *visit* of P to B is a maximal contiguous subpath of P whose vertices lie in B.

Lemma 3.6 (Visit Taxonomy)**.** *For a PC path P and a block B of size s, the visits of P to B come in at most two pieces, classified as follows:*

- *SKIP: zero visits.*
- *ONE: a single visit at a single vertex (a port; an interior vertex only if P has an endpoint there).*
- *FULL: a single visit using all s vertices, contiguously.*
- *PARTIAL$_m$ ($2 \leq m \leq s-1$): a single visit using m vertices, contiguous from one port inward. Requires that an endpoint of P lie in B.*

If neither endpoint of P lies in B, then a two-visit configuration requires both visits to be single-port visits and uses two external edges at each visited port (consuming all four external edges of B). If an endpoint of P lies in B, the corresponding visit may use only one external edge, so the four-external-edge conclusion applies only in the no-endpoint case.

Sketch. Lemma 3.4(b) confines external edges to ports. A visit enters and exits through ports (unless one of P's endpoints is internal to B). Inside B, doubled edges allow any contiguous run to be traversed with alternating colors. A partial visit that stops at an internal vertex requires that vertex to be a path endpoint. Two-visit configurations consume one in-edge and one out-edge per visit at the ports, accounting for the external-edge count above. □

Lemma 3.7 (Block Transducer Parity)**.** *Suppose P visits B once and uses both an entry edge e_{in} and an exit edge e_{out}, of colors c_{in} and c_{out}.*

- *If the visit is ONE, then* $c_{\text{in}} \neq c_{\text{out}}$.
- *If the visit is FULL, then* $c_{\text{in}} \neq c_{\text{out}}$ *iff* s *is odd.*

Proof. A ONE visit at a vertex p uses no internal edges; the PC condition at p requires $c_{\text{in}} \neq c_{\text{out}}$ directly.

A FULL visit uses all $s-1$ internal edges. Label them $e_1, \ldots, e_{s-1}$ in path order. The PC condition at the entry port forces e_1 to have color opposite c_{in}, and color alternation along the chain forces e_j to equal c_{in} iff j is even. Hence $e_{s-1} = c_{\text{in}}$ iff $s-1$ is even iff s is odd. The PC condition at the exit port gives $c_{\text{out}} \neq e_{s-1}$, so $c_{\text{out}} \neq c_{\text{in}}$ iff s is odd. □

The five lemmas specific to σ_m

Proposition 3.8 (Cross-edge structure)**.** *In H_{σ_m}, each block boundary contributes exactly one P-only and one V-only edge. All edges internal to blocks are doubled.*

Sketch. Adjacent blocks in σ_m have $|\sigma(p) - \sigma(p+1)| \neq 1$, so $(p, p+1)$ is P-only at the boundary. The V-only edge at the same boundary comes from the V-successor structure of the maximum-σ value in one block. Internal claim is Lemma 3.4(a). □

Lemma 3.9 (Middle-vertex)**.** *Let B be a 3-block of σ_m with middle vertex p. If a PC path P has p in its interior, then P traverses all three vertices of B contiguously.*

Sketch. p has degree 4 from two doubled pairs (Lemma 3.4(c)). For P to pass through p requires two incident edges of different colors at p; since both pairs are doubled and adjoin the two ports, P uses both port-neighbors of p contiguously with p. □

Lemma 3.10 (Interior 3-block rigidity)**.** *Let B be a 3-block of σ_m both of whose chain-neighbors are singletons. Any PC path P with no endpoint in B uses at most one vertex of B.*

Sketch. Suppose P uses two or three vertices of B without an endpoint there. By Lemma 3.9, if it uses the middle, it uses all three (FULL). For a FULL visit through a 3-block, Lemma 3.7 gives $c_{\text{in}} \neq c_{\text{out}}$, and the singleton neighbors on each side have only one P-neighbor and one V-neighbor in the chain; tracking forced colors at those singletons closes into a 4-cycle, contradicting simplicity. The two-port-vertex case (no middle) similarly closes a forbidden cycle by Lemma 3.6. □

Lemma 3.11 (Block-interval)**.** *For any PC path in H_{σ_m}, the set of visited blocks forms a contiguous interval in the chain.*

Sketch. The block-quotient graph of σ_m is a path. Travelling between non-adjacent blocks requires cross-edges that exist only between chain-adjacent blocks (Proposition 3.8); parity constraints at the singletons rule out skipping a block. □

Lemma 3.12 (End-region bound)**.** *The two-block region consisting of the end 2-block of σ_m and its adjacent extreme 3-block contributes at most 3 vertices to any PC path that extends beyond the region.*

Sketch. Case analysis on the gateway vertex. The 2-block has one external edge and a doubled internal edge, so it contributes at most 2 and only if traversed FULL. The extreme 3-block has one boundary toward the next singleton and one toward the 2-block; Lemma 3.7 on both sides limits its contribution to at most 1 beyond the 2-block, giving the bound of 3. □

Combining the lemmas

For $m \geq 3$, Lemma 3.11 guarantees the visited blocks form an interval. Counting the contributions:

$$\begin{aligned}\rho(\sigma_m) \;\leq\; & \underbrace{3}_{\substack{\text{left end:}\\ \text{Lemma 3.12}}} + \underbrace{3}_{\substack{\text{right end:}\\ \text{Lemma 3.12}}} + \underbrace{(m-2)}_{\substack{\text{interior 3-blocks,}\\ \text{Lemma 3.10}}} + \underbrace{(m-1)}_{\text{singletons}} \\ =\; & 2m+3.\end{aligned}$$

For $m = 1$, σ_1 has three blocks (composition $(2, 3, 2)$) and the budget collapses to $\rho \leq 2 + 3 = 5$. For $m = 2$, σ_2 has one interior singleton between two 3-blocks (composition $(2, 3, 1, 3, 2)$), and a direct check gives $\rho \leq 7$.

The lemma sketches above are roadmap-level rather than fully expanded. Full proofs require additional case analysis at the boundaries and are available in the project archive.

3.2 Exhaustive verification through $n = 14$

A C++ verifier with OpenMP enumerates S_n and computes $\rho(\sigma)$ for each σ via a depth-first search over alternating-color extensions with backtracking. The full S_{14} run took approximately 5 days of wall-clock time on a 16-core workstation. No symmetry reduction was applied in the main verifier; symmetry-quotient codes were used only for spot-checking specific orbits, and two independent enumerators agree on the values below.

n	$\lvert S_n\rvert$	$\rho_{\min}(n)$	$\lceil (n+3)/2 \rceil$	comparison
≤ 6	≤ 720	n	—	$\rho_{\min} = n$, every σ Hamiltonian
7	5,040	5	5	tight
8	40,320	6	6	tight
9	362,880	7	6	+1
10	3,628,800	8	7	+1
11	39,916,800	7	7	tight
12	479,001,600	8	8	tight
13	6,227,020,800	9	8	+1
14	87,178,291,200	10	9	+1

The bound is tight at $n = 7, 8, 11, 12$ and strictly slack elsewhere. At $n = 4m + 3$ the extremal family σ_m realizes the lower bound. As shown next, σ_m is not globally extremal once stacking is allowed.

4 Refutation: counterexamples to the half-bound conjecture

4.1 An $n = 18$ counterexample

Let $\sigma^* = (1, 0, 4, 3, 2, 5, 8, 7, 6, 11, 10, 9, 12, 15, 14, 13, 17, 16) \in S_{18}$. Its block structure has composition $(2, 3, 1, 3, 3, 1, 3, 2)$ with $b(\sigma^*) = 8$ blocks:

$$B_1{=}\{0,1\},\ B_2{=}\{2,3,4\},\ B_3{=}\{5\},\ B_4{=}\{6,7,8\},\ B_5{=}\{9,10,11\},\ B_6{=}\{12\},\ B_7{=}\{13,14,15\},\ B_8{=}\{16,17\}.$$

Theorem 4.1. $\rho(\sigma^*) = 10 < 11 = \lceil (18 + 3)/2 \rceil$. *Conjecture 3.1 fails at $n = 18$.*

Proof. Lower bound. The sequence

$$\Pi^* \;=\; (13,\ 14,\ 15,\ 12,\ 11,\ 6,\ 5,\ 2,\ 1,\ 0)$$

is a PC path on 10 vertices in H_{σ^*}. Direct verification of each edge: $\{13, 14\}$ and $\{14, 15\}$ are doubled (internal to B_7); $\{12, 15\}$ is V-only (V-edge between B_7 and B_6); $\{11, 12\}$ is P-only; $\{6, 11\}$ is V-only (since $\sigma^{*,-1}(8) = 6$ and $\sigma^{*,-1}(9) = 11$); $\{5, 6\}$ is P-only; $\{2, 5\}$ is V-only (V-edge between B_2 and B_3); $\{1, 2\}$ is P-only; $\{0, 1\}$ is doubled. Assigning colors $V, P, V, P, V, P, V, P, V$ to the nine edges produces a valid PC traversal.

Upper bound. An exact bitmask dynamic program with state (v, c, S) – current vertex, last edge color, set of visited vertices – computes the maximum number of additional vertices reachable along a color-$\neq c$ edge to a vertex outside S. Running this from every starting vertex and both initial-color conventions returns $\rho(\sigma^*) = 10$. A separate depth-first backtracking enumerator with the same edge set gives the same value. □

4.2 How the counterexample was found

The construction is a single-position deletion from $\sigma_4 \in S_{19}$ (composition $(2, 3, 1, 3, 1, 3, 1, 3, 2)$, $\rho(\sigma_4) = 11$). Delete the center singleton (position 9, value 9) and relabel the remaining σ-values down by 1 for positions past the deletion:

$$\sigma_4 \;=\; (1, 0, 4, 3, 2, 5, 8, 7, 6, \underline{9}, 12, 11, 10, 13, 16, 15, 14, 18, 17) \;\longrightarrow\; \sigma^*.$$

Removing the singleton fuses two flanking 3-blocks across a non-± 1 gap, changing the composition from $(2, 3, 1, 3, 1, 3, 1, 3, 2)$ to $(2, 3, 1, 3, 3, 1, 3, 2)$. The block count drops from 9 to 8. The deficit $d = n - \rho$ stays at 8: the singleton contributed exactly one vertex to the optimal path of σ_4, so removing it reduces both n and ρ by 1.

This deletion was implicit in earlier numerical data, recorded at the level of the deficit without spelling out the implication. Re-examining: σ^* has $b = 8$ and $d = 8$, so $d > b - 1 = 7$. Since the per-block deficit bound $d \leq b - 1$ (verified through $n \leq 12$) had been treated as a candidate route to a linear bound, the conflict was diagnostic, and a recomputation of $\rho(\sigma^*)$ produced the counterexample.

4.3 Stacking: an asymptotic family

The deletion construction iterates. For a module $M = (s_1, s_2, \ldots, s_r)$ of block sizes summing to $|M|$, define $\sigma^{(k)}$ at $n_k = k \cdot |M| + 7$ by taking blocks of sizes $(2) \,\|\, M^k \,\|\, (3, 2)$ and reversing each. For $M = (3, 1, 3, 3, 1)$ with $|M| = 11$, direct computation gives:

k	n_k	b	$\rho(\sigma^{(k)})$	$d = n - \rho$	ρ/n
0	7	3	5	2	0.7143
1	18	8	10	8	0.5556
2	29	13	15	14	0.5172
3	40	18	20	20	0.5000
4	51	23	25	26	0.4902

The slope $\Delta\rho/\Delta n = 5/11 \approx 0.4545$ is consistent across all five tabulated values. C++ enumeration at $n = 18, 29, 40$ confirms exact ρ; the $n = 51$ row used a bitmask DP spot-checked against the exhaustive search at smaller sizes.

A separate module search up to size 28 found $M' = (3,3,3,3,3,1,3,3,3,3)$ achieving slope $10/28 \approx 0.357$ at $k = 1$. The corresponding stacking statement at $k = 1$ is verified directly; the asymptotic claim is again conjectural.

Conjecture 4.2 (Stacking, $M = (3,1,3,3,1)$)**.** *For every $k \geq 0$, $\rho(\sigma^{(k)}) = 5k + 5$. Equivalently, $c_* \leq 5/11$.*

Conjecture 4.3 (Stacking, $M' = (3,3,3,3,3,1,3,3,3,3)$)**.** *For every $k \geq 0$, the analogous family built from M' has ρ growing at slope $10/28$. Equivalently, $c_* \leq 10/28$.*

The conjectures are open. A direct verification at finitely many k does not by itself prove an upper bound on c_*. Heuristically, the optimal PC path in $\sigma^{(k)}$ appears to traverse one full 3-block per period and pick up one singleton, repeating the gadget structure of the $k = 1$ case; a proof would require a stacking lemma controlling the interaction between consecutive periods.

What is unconditional: the single permutation σ^* at $n = 18$ refutes Conjecture 3.1 outright. Only the asymptotic upper bounds on c_* remain conjectural.

5 What survives

The refutation leaves several structural results intact. The most useful are an unconditional bound from the largest block, an extension by one, and a conditional lifting from a quotient permutation.

5.1 Unconditional bounds

Theorem 5.1 (Largest-block bound)**.** *For every $\sigma \in S_n$, $\rho(\sigma) \geq s_{\max}(\sigma)$.*

Proof. Let B be a block of maximum size $s_{\max}$. By Lemma 3.4(a), every internal edge of B is doubled; traversing B in order with alternating P, V across consecutive edges gives a PC path on $s_{\max}$ vertices. □

Theorem 5.2 (One-vertex extension)**.** *If $b(\sigma) \geq 2$, then $\rho(\sigma) \geq s_{\max}(\sigma) + 1$.*

Proof. If $s_{\max} = 1$, all blocks are singletons; with $b \geq 2$ the P-edge $\{0,1\}$ exists and gives a PC path on $2 = s_{\max} + 1$ vertices.

Otherwise $s_{\max} \geq 2$. Let B be a maximum block and $p \in \{p_\ell, p_r\}$ a port at which B has an external edge e of color c (such a port exists because $b \geq 2$). Traverse B from the opposite port toward p, choosing the starting color so the final internal edge has color $\neq c$. The PC condition at p then permits exiting along e to a vertex outside B, giving a PC path on $s_{\max} + 1$ vertices. □

These bounds are weak in absolute terms: the stacked family from Section 4.3 has $s_{\max} = 3$ regardless of n, so Theorem 5.2 alone gives only $\rho \geq 4$ there. They serve as a floor that any combined bound must clear.

5.2 Conditional quotient lifting

Definition 5.3. For $\sigma \in S_n$ with $b(\sigma) = b$ blocks $B_1, \ldots, B_b$ listed in position order, the *block-quotient permutation* $\tau_\sigma \in S_b$ is obtained by ranking the blocks by their σ-value range. Each block of σ has a single contiguous range of σ-values (since blocks are ± 1-runs); τ_σ maps the position-rank of a block to its σ-rank. The *quotient graph* H_{τ_σ} is built from τ_σ as in Definition 2.2, with one vertex per block.

Example 5.4. For $\sigma = (0, 2, 3, 1, 4) \in S_5$, the position-ordered blocks are $\{0\}, \{1, 2\}, \{3\}, \{4\}$ with σ-value ranges $[0, 0], [2, 3], [1, 1], [4, 4]$. Sorting by σ-value gives the order $\{0\}, \{3\}, \{1, 2\}, \{4\}$, so $\tau_\sigma \in S_4$ maps $0 \mapsto 0$, $1 \mapsto 2$, $2 \mapsto 1$, $3 \mapsto 3$, i.e., $\tau_\sigma = (0, 2, 1, 3)$.

Empirically, $\rho(\sigma) \geq \rho(\tau_\sigma)$ for all σ tested ($n \leq 11$). A path-by-path *lift* (expand each block-visit in a quotient PC path into a corresponding visit in H_σ) is not always possible: by Lemma 3.7, if a quotient PC path demands a FULL traversal of an internal block of even size, the lift forces $c_{\text{in}} = c_{\text{out}}$, contradicting the PC condition at the corresponding block-vertex in the quotient.

The smallest witness is the σ from Example 5.4: a maximum PC path in H_{τ_σ} that visits $\{2, 3\}$ as a non-endpoint block requires a FULL transit through this size-2 block. The naive lift to H_σ then fails the parity check. Note that this only shows the *naive* lift fails: the inequality $\rho(\sigma) \geq \rho(\tau_\sigma)$ still holds here, since $\rho(\sigma) = 5 \geq \rho(\tau_\sigma) = 4$, but no general method for producing a non-naive lift is known.

Theorem 5.5 (Odd-only quotient lift)**.** *Call a block of σ* internal *if it is neither the first nor the last in the block chain. If every internal block of σ has odd size, then $\rho(\sigma) \geq \rho(\tau_\sigma)$.*

Sketch. Take a maximum PC path Q in H_{τ_σ} and lift it to H_σ by expanding each block visit. Internal-block visits in Q correspond to either same-port (ONE) or different-port (FULL) traversals in H_σ. By hypothesis every internal block has odd size, so Lemma 3.7 gives $c_{\text{in}} \neq c_{\text{out}}$ for every FULL lift, matching the PC condition in Q. End blocks are not constrained the same way because they carry only one external edge, with no parity match to verify. □

The hypothesis is satisfied by the original extremal family σ_m, whose internal blocks have sizes 3 and 1. The general inequality $\rho(\sigma) \geq \rho(\tau_\sigma)$ remains conjectural.

6 Landscape under adjacent transpositions

Independent evidence for the local extremality of σ_m comes from the structure of ρ as a function on S_n under adjacent transpositions. View S_n as a graph with $\sigma \sim \sigma'$ when σ' differs from σ by swapping two consecutive positions. The eight-element symmetry group generated by reversing P, reversing V, and swapping the two color classes acts on S_n and preserves ρ; we count *canonical* local minima by taking the lexicographically least representative in each orbit.

n	$\rho_{\min}(n)$	canonical local minima at $\rho \leq \rho_{\min} + 1$	global min isolated?
7	5	2	yes
8	6	5	partially
9	7	118	no (flat landscape)
10	8	695	no (flat landscape)
11	7	3	yes (every neighbor strictly higher)

At $n = 4m+3$ with $m \geq 2$, the global minimum is an isolated well: every adjacent transposition from σ_m strictly increases ρ. At other n, the landscape is flat with hundreds of local minima. This was the strongest empirical signal for an exchange-style proof of the half-bound conjecture, the idea being that σ_m is the unique global minimizer at $n = 4m + 3$. The refutation in Section 4 reframes this evidence: the isolated-well structure of σ_m inside S_{4m+3} is compatible with the existence of better constructions at other values of n, which is what the stacked family provides.

7 Current state

Proved unconditionally.

- Conversion Lemma (Theorem 2.5): ρ-lower bounds transfer to $f(n)$.
- Local block calculus: Port Structure (Lemma 3.4), Visit Taxonomy (Lemma 3.6), Block Transducer Parity (Lemma 3.7).
- Extremal value: $\rho(\sigma_m) = 2m + 3$ at $n = 4m + 3$ (Theorem 3.2). The proof above is at the sketch level for the five σ_m-specific lemmas; full proofs are available in the project archive.
- Largest-block bound and one-vertex extension (Theorems 5.1, 5.2).
- Odd-only quotient lift (Theorem 5.5).

Disproved unconditionally.

- The half-bound conjecture (Conjecture 3.1) is false at $n = 18$ (Theorem 4.1).
- The deficit bound $d(\sigma) \leq b(\sigma) - 1$, consistent with all data through $n \leq 12$, fails at σ^*.
- Naive path-by-path quotient lifting (Section 5.2).

Computational evidence, conjecturally true.

- Stacking conjecture for $M = (3, 1, 3, 3, 1)$ (Conjecture 4.2): exact values verified for $k \leq 4$ (slope 5/11); the asymptotic claim $c_* \leq 5/11$ depends on the stacking persisting.
- Stacking conjecture for $M' = (3, 3, 3, 3, 3, 1, 3, 3, 3, 3)$ (Conjecture 4.3): verified at $k = 1$ (slope 10/28); $c_* \leq 10/28$ depends on persistence.

Open.

- Is $c_* > 0$? Equivalently, is there a constant $c > 0$ such that $\rho(\sigma) \geq cn - O(1)$ for all σ? This is the central question for the Lovász application.
- The full quotient-lift inequality $\rho(\sigma) \geq \rho(\tau_\sigma)$, verified empirically for $n \leq 11$, without the odd-block hypothesis.
- The two stacking conjectures above.

Computational state. $\rho_{\min}(n)$ is verified for $n \leq 14$ (over 8.7×10^{10} permutations). The smallest known counterexample to Conjecture 3.1 is σ^* at $n = 18$; the values $n = 15, 16, 17$ have not been exhaustively checked, so we do not claim $n = 18$ is the smallest counterexample value. (At the current rate, $n = 15$ would take roughly $15\times$ the wall-clock time of $n = 14$, putting it out of reach without substantial compute.) Random sampling at $n = 20$ (uniform on S_{20}, 10^6 trials) found no permutation with deficit greater than 3, suggesting the bad permutations are sparse, though no quantitative rarity claim is established.

A clean self-contained question for readers who do not care about the Lovász application:

> *Does there exist $c > 0$ such that every two-edge-colored multigraph that is the union of two Hamiltonian paths on the same vertex set contains a properly colored simple path on at least cn vertices?*

A positive answer with any explicit $c > 0$ would already improve the Lovász bound. Conjecture 4.2, if true, gives the upper bound $c \leq 5/11$.

8 Discussion

The structural picture is clean even after the refutation. The hard combinatorics lives at block interfaces, where doubled-edge shortcuts disappear and parity conflicts between the two color classes accumulate. The local calculus in Section 3.1 gives a complete account of how a PC path can interact with a single block. The remaining difficulty is global: how interfaces compose along the chain. Within the local analysis developed here, σ_m is a locally optimal arrangement of 3-blocks and singletons; the singleton-deletion family is a globally more efficient arrangement that the σ_m analysis does not see.

Natural next steps are to search the module space more thoroughly for asymptotic minimizers, to extend exhaustive verification toward $n = 18$ (which would require either a substantial compute budget or symmetry-aware enumeration), and to attack the unconditional quotient-lift inequality $\rho(\sigma) \geq \rho(\tau_\sigma)$. A linear bound on ρ now appears to require either a forbidden-substructure characterization of permutations with small ρ/n, or a structural theorem on alternating paths in unions of two Hamiltonian paths that goes beyond the cited literature.

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